

Conda Environment: Exporting & Recreating

1 What is Conda?

Conda is a *package and environment manager* used in Python (and ML projects) to:

- Create **isolated environments**
- Manage **packages and dependencies**
- Ensure **version consistency** across devices and teammates

Think of it as a “sandbox” that keeps your ML libraries separate — so your TensorFlow version doesn’t break PyTorch, etc.

2 Why Use Conda in MLOps?

In MLOps, different teams (dev, staging, prod) must run the **same versions** of libraries.

Conda helps by:

- Capturing environment details (`.yaml` file)
- Rebuilding the same environment anywhere
- Preventing “works on my machine” issues

Example:

You trained a model on your laptop with Python 3.11, NumPy 1.26, and Pandas 2.2

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Conda can export these versions so your teammate or CI/CD pipeline uses *exactly* the same setup.

3 Conda Environment Basics

Command	Description
<code>conda create -n env_name python=3.11</code>	Create a new environment
<code>conda activate env_name</code>	Activate the environment
<code>conda deactivate</code>	Deactivate it
<code>conda env list</code>	List all conda environments
<code>conda remove -n env_name --all</code>	Delete environment

4 Exporting an Environment (creating YAML file)

When your environment is ready (with all installed packages):

```
conda env export > environment.yml
```

This command:

- Creates a **YAML file**
- Stores all package names, versions, and channels
- Makes it easy for others to clone your setup

Example output:

```
name: mlops-env
dependencies:
- python=3.11
- numpy=1.26.4
- pandas=2.2.1
- scikit-learn=1.4.2
- pip:
```

- fastapi==0.115.0

5 Recreating an Environment (from YAML file)

To create the same environment elsewhere:

```
conda env create -f environment.yml
```

✓ Conda reads the file and installs everything exactly as before.

6 Updating an Existing Environment

If the `environment.yml` changes (you added a new library):

```
conda env update -f environment.yml --prune
```

- `--prune` removes packages not listed in the file (keeps it clean).
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7 Export Without Build Numbers (cleaner YAML)

Sometimes you don't want OS-specific or build version info (for portability):

```
conda env export --no-builds > environment.yml
```

This makes it easier to recreate across Windows, Linux, and macOS.

8 Environment Location and Sharing

- Default location → `C:\Users\<user>\anaconda3\envs\<env_name>` (Windows)

- You can send `environment.yml` via GitHub or email — it's just a text file.

9 Conda vs requirements.txt

Conda (YAML)	Pip (requirements.txt)
Includes channels + dependencies	Only package names
Supports multiple languages	Python only
Heavier but reproducible	Simpler but less strict

In MLOps, we prefer **Conda YAML** for complex setups (e.g., MLflow, FastAPI, PyTorch, etc.).

10 Quick Recap

Task	Command
Create environment	<code>conda create -n env_name python=3.11</code>
Activate environment	<code>conda activate env_name</code>
Export environment	<code>conda env export > environment.yml</code>
Recreate environment	<code>conda env create -f environment.yml</code>
Update environment	<code>conda env update -f environment.yml --prune</code>
Clean export (no builds)	<code>conda env export --no-builds > environment.yml</code>

15 MCQs (with answers + explanations)

1 Which command creates a new Conda environment?

→ `conda create -n env_name python=3.11`

- 2 Which command activates an environment?
→ `conda activate env_name`
 - 3 Which file type is generated during export?
→ YAML file (`environment.yml`)
 - 4 Command to export current environment?
→ `conda env export > environment.yml`
 - 5 Purpose of `environment.yml`?
→ To record and recreate all dependencies.
 - 6 Command to recreate environment from file?
→ `conda env create -f environment.yml`
 - 7 To update existing environment from YAML:
→ `conda env update -f environment.yml --prune`
 - 8 What does `--no-builds` flag do?
→ Exports without OS-specific build info.
 - 9 To list all environments:
→ `conda env list`
 - 10 To delete an environment completely:
→ `conda remove -n env_name --all`
 - 11 Conda YAML file includes:
→ Channels, package names, and versions.
 - 12 Difference between `pip install -r` and Conda YAML?
→ Conda YAML includes more metadata and supports non-Python dependencies.
 - 13 Which command deactivates current environment?
→ `conda deactivate`
 - 14 Default name of export file?
→ `environment.yml`
 - 15 Why use Conda environments in MLOps?
→ To ensure consistency and reproducibility across training, testing, and deployment stages.
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10 Short Answer Questions (2–3 sentences each)

1 What is Conda used for?

To manage isolated environments and packages so ML projects have consistent dependencies.

2 Why is environment reproducibility important?

Because if the same code runs on different library versions, results may differ or fail.

3 How do you export a Conda environment?

Run `conda env export > environment.yml` to generate a YAML file listing all packages.

4 How do you recreate an environment from YAML?

Run `conda env create -f environment.yml`.

5 What does the `--prune` flag do in update command?

It removes packages not listed in the YAML file.

6 How is Conda better than pip for MLOps?

Conda can manage system-level libraries and multiple languages, while pip only handles Python packages.

7 What is inside `environment.yml`?

The environment name, Python version, library versions, and pip dependencies.

8 How do you remove an environment completely?

Run `conda remove -n env_name --all`.

9 When should you use `--no-builds`?

When you want a portable environment across different operating systems.

10 Where is the `environment.yml` used in CI/CD?

In the build stage — the pipeline uses it to install the exact dependencies before testing or deployment.